

Week 4: Measuring and Hedging FX Exposure

FINA3020: International Finance

Henry Zhang

CUHK Business School

Term 1, 2026–2027

- **Part I: Measuring FX Exposure**

- Defining exposure: Transaction, Economic, and Translation.
- Identifying direction, timing, and netting across currencies.

- **Part II: Transaction Hedges and FX Swaps**

- Unhedged, forward, money market, and option hedges.
- FX swaps: matching cash-flow dates and converting loan currencies.
- CIP pricing and comparison of hedged borrowing costs.

- **Part III: Payoff Diagrams & Strategic CFO Choices**

- Visual payoff comparisons and break-even analysis.
- In-Class Game Round 4: *CFO Hedge Meeting*.

- **Part IV: Economic & Translation Exposure Overview**

- Operating cash-flow exposure vs accounting translation/remeasurement effects.

[View Exposure Summary Table →](#)

Foreign exchange exposure measures the **sensitivity** of a firm's home-currency cash flows, profits, or market value to exchange rate movements.

1. Transaction

Sensitivity of **contractual, committed** foreign currency cash flows due at a known future date T .

Focus of today's hedging tools.

2. Economic

Sensitivity of **future operational** cash flows and long-term competitiveness to real exchange rates.

Affects even purely domestic firms!

3. Translation

Sensitivity of **consolidated accounting** statements to translating or remeasuring foreign operations into the reporting currency.

Often affects equity or earnings.

[Detailed Comparison Table →](#)

Step 1: Identifying Direction & Timing of Exposure

[Core Concept]

Before choosing any hedging instrument, a CFO must identify:

- ① *Currency pair*: Base foreign currency $\mathcal{F}$ vs quote/home currency H, written $S_{\mathcal{F}/H}$ and measured in H per unit of $\mathcal{F}$.
- ② *Direction of exposure*: Is the firm receiving or paying $\mathcal{F}$?
- ③ *Timing*: Exact settlement date T .

Position	Contractual Nature	FX Risk Direction	Hedging Goal
Foreign Receivable	Inflow of $\mathcal{F}$ at date T (Asset)	$\mathcal{F}$ depreciates ($S_{\mathcal{F}/H} \downarrow$) $\implies$ Fewer H received	Establish a floor on revenue
Foreign Payable	Outflow of $\mathcal{F}$ at date T (Liability)	$\mathcal{F}$ appreciates ($S_{\mathcal{F}/H} \uparrow$) $\implies$ More H required	Establish a cap on cost

Never choose a hedge before establishing exposure direction. A foreign-currency receivable is hedged with a put; buying a call fails to protect its downside and adds a separate option position.

Receivable or Payable: Follow EUR 100

Illustrative one-year cash flows. The quote is **HKD per EUR**: $S_0 = 8.50$.

- ① **Receivable:** a customer owes us EUR 100. At the spot rate, the value is $100 \times 8.50 = 850$ HKD.
 - If the rate falls to 8.00, we receive $100 \times 8.00 = 800$ HKD. The change is $800 - 850 = -50$ HKD. **A weaker EUR hurts.**

- ② **Payable:** we owe a supplier EUR 100. If the rate rises to 9.00, buying the EUR costs $100 \times 9.00 = 900$ HKD.
 - The cost rises by $900 - 850 = 50$ HKD. **A stronger EUR hurts.** Sell EUR forward for a receivable, buy EUR forward for a payable.

Receivable or Payable: Follow EUR 100

Illustrative one-year cash flows. The quote is **HKD per EUR**: $S_0 = 8.50$.

- ① **Receivable:** a customer owes us EUR 100. At the spot rate, the value is $100 \times 8.50 = 850$ HKD.
 - If the rate falls to 8.00, we receive $100 \times 8.00 = 800$ HKD. The change is $800 - 850 = -50$ HKD. **A weaker EUR hurts.**

- ② **Payable:** we owe a supplier EUR 100. If the rate rises to 9.00, buying the EUR costs $100 \times 9.00 = 900$ HKD.
 - The cost rises by $900 - 850 = 50$ HKD. **A stronger EUR hurts.** Sell EUR forward for a receivable, buy EUR forward for a payable.

Multinational firms rarely hedge individual transactions separately. Instead, they aggregate inflows and outflows per currency to calculate **net exposure**.

Foreign Currency ($\mathcal{F}$)	Inflow	Outflow	Net $\mathcal{F}$ Exposure	Spot $S_{\mathcal{F}/\text{USD}}$	Net USD Exposure
EUR (€)	€8M	€3M	+€5M (Long)	\$1.10	+\$5.5M
GBP (£)	£4M	£6M	-£2M (Short)	\$1.30	-\$2.6M
JPY (¥)	¥500M	¥500M	¥0 (Matched)	\$0.007	\$0

- **Multilateral Netting:** Matching receivables and payables in the same currency and settlement window reduces the amount externally hedged; legal-entity, timing, tax, and settlement constraints can limit netting.
- **Cross-Currency Portfolio Effect:** If EUR and GBP move together against USD, a long EUR position partially offsets a short GBP position.

Full Netting Worked Example →

Netting: Calculate Each Currency Separately

- ① **EUR:** receipts EUR 8M minus payments EUR 3M leave $8 - 3 = +5$ million EUR. At 1.10 USD per EUR, $5 \times 1.10 = +5.5$ million USD.
- ② **GBP:** receipts GBP 4M minus payments GBP 6M leave $4 - 6 = -2$ million GBP. At 1.30 USD per GBP, $-2 \times 1.30 = -2.6$ million USD.
- ③ **JPY:** $500 - 500 = 0$ million JPY. Matching the currency *and settlement date* removes this contractual net exposure.
- ④ The USD values sum to $5.5 - 2.6 = 2.9$ million. But EUR and GBP are different currencies: keep both exposures when measuring risk or choosing hedges.

- For each currency, measure spot prices against the same home currency and compute $R_t = S_t/S_{t-1} - 1$. Exchange-rate levels and returns answer different questions.
- Sample variance is $s_X^2 = \sum_t (R_{X,t} - \bar{R}_X)^2 / (T - 1)$. Standard deviation has the units of returns. Variance has squared units.
- Covariance is $s_{XY} = \sum_t (R_{X,t} - \bar{R}_X)(R_{Y,t} - \bar{R}_Y) / (T - 1)$. Correlation is $\rho_{XY} = s_{XY} / (s_X s_Y)$.
- In a return scatterplot, positive correlation $\iff$ same-signed movements tend to occur together. Correlation measures linear association and does not establish causation.

From currency prices to a return scatterplot

Both exchange-rate series must use the same dates and return definition. Hypothetical example:

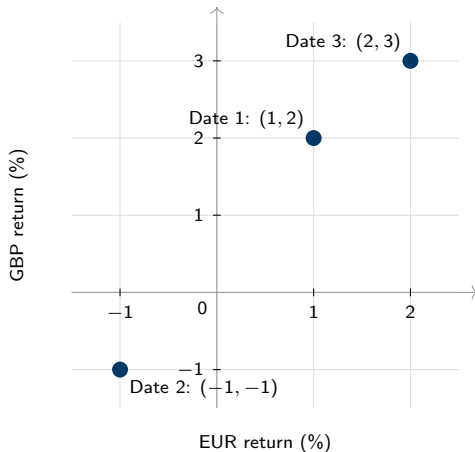
Date	EUR/USD	GBP/USD	EUR return	GBP return
0	1.1000	1.2500	—	—
1	1.1110	1.2750	1.0%	2.0%
2	1.09989	1.26225	-1.0%	-1.0%
3	1.1218878	1.3001175	2.0%	3.0%

- A return scatterplot has points (1, 2), (−1, −1), and (2, 3) in percentage points. Its sample correlation is about 0.996.
- Volatility can differ: the sample standard deviations are 1.528 and 2.082 percentage points. Common trends in price levels can be misleading.

Return calculations →

Correlation calculations →

Currency return scatterplot



Date	EUR/USD	GBP/USD	EUR return	GBP return
0	1.1000	1.2500	—	—
1	1.1110	1.2750	1.0%	2.0%
2	1.09989	1.26225	-1.0%	-1.0%
3	1.1218878	1.3001175	2.0%	3.0%

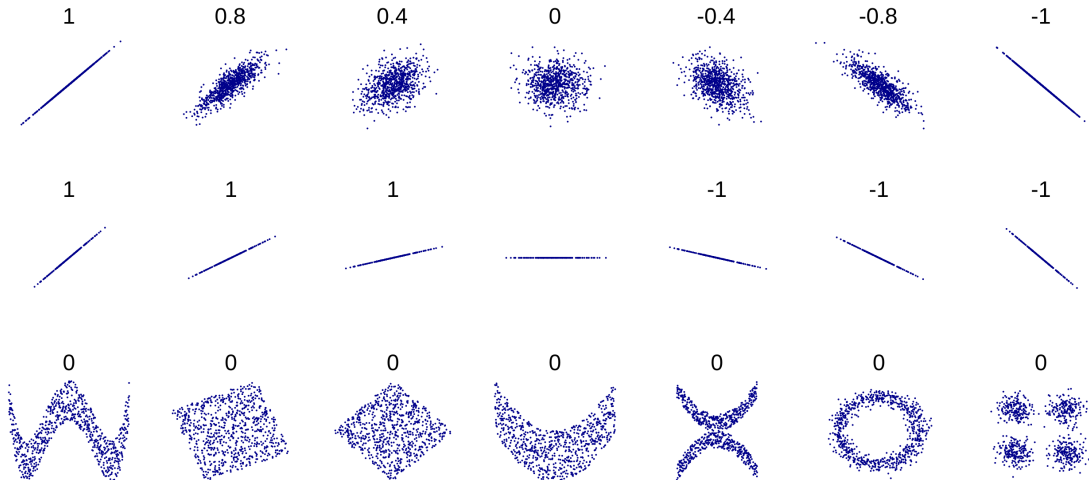
Each point pairs the two currency returns on the same date.

Sample correlation: $\rho \approx 0.996$.

The returns move closely together, although GBP returns vary more.

Hypothetical data from the preceding slide.

What do correlations represent?



FX Correlations Example

Correlations	EUR	GBP	AUD	NZD	JPY	CNY	CHF	CAD	MXN	INR	BRL	RUB	KRW	USD
EUR		.65	.82	-.03	-.45	.70	.97	.36	.87	-.64	.80	-.02	.29	.95
GBP	.65		.65	.47	.08	.57	.57	.63	.53	-.11	.52	.34	.70	.79
AUD	.82	.65		.01	-.48	.86	.80	.28	.87	-.60	.83	-.08	.39	.79
NZD	-.03	.47	.01		.63	-.21	-.07	.81	-.25	.65	-.31	.57	.63	.20
JPY	-.45	.08	-.48	.63		-.61	-.46	.31	-.72	.76	-.65	.16	.15	-.18
CNY	.70	.57	.86	-.21	-.61		.67	.14	.87	-.68	.85	-.02	.44	.63
CHF	.97	.57	.80	-.07	-.46	.67		.31	.84	-.63	.73	-.06	.24	.91
CAD	.36	.63	.28	.81	.31	.14	.31		.16	.31	.05	.65	.75	.53
MXN	.87	.53	.87	-.25	-.72	.87	.84	.16		-.78	.92	-.03	.27	.75
INR	-.64	-.11	-.60	.65	.76	-.68	-.63	.31	-.78		-.80	.43	.15	-.45
BRL	.80	.52	.83	-.31	-.65	.85	.73	.05	.92	-.80		-.17	.20	.69
RUB	-.02	.34	-.08	.57	.16	-.02	-.06	.65	-.03	.43	-.17		.64	.07
KRW	.29	.70	.39	.63	.15	.44	.24	.75	.27	.15	.20	.64		.43
USD	.95	.79	.79	.20	-.18	.63	.91	.53	.75	-.45	.69	.07	.43	

Convert each net foreign-currency cash flow into the same home currency at current spot. Let x_i be that signed home-currency amount.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

- For positive asset positions, value weights satisfy $w_i = x_i / \sum_j x_j$ and $\sigma_p^2 = w^\top \Sigma w$.
- A payable is a negative exposure. Positive currency correlation can offset a receivable and a payable, while it reinforces two receivables.
- If net portfolio value is near zero, report home-currency profit-and-loss risk using $x^\top \Sigma x$. Dividing by a tiny net value produces misleading weights.

Signed Exposures

Illustration: a EUR receipt is worth USD 100 today and a GBP payment is worth USD 100. Both currencies have 10% volatility and correlation +1.

- ① Keep the signs: $x_{\text{EUR}} = +100$, $x_{\text{GBP}} = -100$.
- ② If both currencies rise by 10%, the receipt gains $100(0.10) = 10$ USD. The payment becomes $100(0.10) = 10$ USD more expensive.
- ③ Net change: $+10 - 10 = 0$ USD. In the variance formula,

$$100^2(0.1)^2 + (-100)^2(0.1)^2 + 2(100)(-100)(1)(0.1)(0.1) = 0.$$

- ④ With *two receipts*, the cross term is positive. Variance is $100 + 100 + 200 = 400$, so standard deviation is USD 20.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

A 95% one-day VaR is a loss threshold exceeded with 5% probability under the model.

$$\text{VaR}_{95\%} = 1.645\sigma_{\Delta V} - E[\Delta V]$$

- Hypothetical receivable: MXN 10M, MXN/USD = 0.09, one-day currency volatility 1%, and mean return zero.
- Current value is USD 900,000. VaR is $900,000(1.645)(0.01) = \text{USD } 14,805$.
- The 5th-percentile rate is $0.09(1 - 0.01645) = 0.0885195$, giving value USD 885,195. Losses can exceed USD 14,805.

Note that tail risk losses may have much lower than 5% probability, and might be far higher than the VaR threshold.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

Single-Currency VaR: Keep Track of the Units

- ① **Value today:** $10,000,000 \text{ MXN} \times 0.09 \text{ USD per MXN} = 900,000 \text{ USD}$.
- ② **One-day dollar standard deviation:** $900,000 \times 0.01 = 9,000 \text{ USD}$. The 1% volatility is a decimal of 0.01.
- ③ **95% normal-model VaR, zero mean:** $1.645 \times 9,000 = 14,805 \text{ USD}$. The 1.645 multiplier is the normal 95th-percentile cutoff.
- ④ **Value at that loss threshold:** $900,000 - 14,805 = 885,195 \text{ USD}$. Divide by MXN 10M to get 0.0885195 USD per MXN.
- ⑤ Under this model, the loss exceeds USD 14,805 on 5% of days. This is **not a maximum loss**.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

Suppose IDR and THB receivables have current USD values of 600,000 and 400,000. Monthly volatilities are 7% and 8%, with correlation 0.5.

$$\sigma_p = \sqrt{0.6^2(0.07)^2 + 0.4^2(0.08)^2 + 2(0.6)(0.4)(0.07)(0.08)(0.5)} = 6.428\%.$$

$$\text{VaR}_{95\%} = 1.645(0.06428)(\text{USD } 1\text{M}) \approx \text{USD } 105,742.$$

- This normal-return example has less risk than either currency held alone. Diversification depends on the weights and correlation.
- VaR itself does not require normality. Historical simulation and other models give different estimates. Expected shortfall measures the average loss beyond the VaR threshold.
- Stress tests should change volatility, correlation, convertibility, and cash-flow amounts. Historical estimates can miss a joint currency crash.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

Portfolio VaR: Calculate the Three Variance Terms

Continuing the IDR and THB example, compute the total current value: USD
 $600,000 + 400,000 = 1,000,000$.

The weights are 0.6 and 0.4. Use monthly volatilities 0.07 and 0.08.

- ① First currency: $0.6^2 \times 0.07^2 = 0.001764$.
- ② Second currency: $0.4^2 \times 0.08^2 = 0.001024$.
- ③ Cross term: $2 \times 0.6 \times 0.4 \times 0.07 \times 0.08 \times 0.5 = 0.001344$.
- ④ Add, then take the square root: $\sigma_p = \sqrt{0.004132} = 0.0642806$, or 6.428% per month.
- ⑤ Assuming normal returns and zero mean, monthly VaR is
 $1.645 \times 0.0642806 \times 1,000,000 \approx 105,742$ USD.

$$\sigma_{\Delta V}^2 = x_X^2 \sigma_X^2 + x_Y^2 \sigma_Y^2 + 2x_X x_Y \rho_{XY} \sigma_X \sigma_Y.$$

A firm facing a foreign receivable or payable has 4 baseline choices:

- ① **Unhedged (Spot at T):** Do nothing today; convert cash flows at the future spot rate $S_{\mathcal{F}/H,T}$. Exposes the home-currency value to FX movements.
- ② **Forward Contract Hedge:** Lock in $F_{\mathcal{F}/H,0,T}$ today. For a certain amount and date, this fixes the conversion rate, but leaves counterparty, liquidity, and settlement risks; a new at-market forward normally has zero initial value, not zero economic cost.
- ③ **Money Market Hedge (MMH):** Replicate the forward contract by borrowing in one currency, converting at spot S_0 , and investing in the other currency.
- ④ **Currency Option Hedge:** Purchase a **call** (for payables) or **put** (for receivables). Pays an upfront “insurance” premium p for asymmetric protection.

Commodity markets →

Hedge policy and practical alternatives

- A full hedge prioritizes cash-flow stability. A selective hedge varies the hedge ratio with exposure certainty, risk limits, cost, and market views.
- Leading or lagging shifts payment dates if contracts and counterparties permit. Financing costs and supplier relationships matter.
- A proxy hedge uses a correlated currency when the exact hedge is unavailable. Residual basis risk remains, especially if correlation changes.
- Natural hedges match receipts and payments by currency and date. Diversifying currencies helps only if their net exposures offset in relevant scenarios.

[Tulip contracts →](#)

[Tulip price exhibit →](#)

Consider a US exporter expecting a receivable of $\mathcal{F}$ 1,000,000 at time T .

1. Unhedged Strategy

Payoff at T (in USD):

$$\text{Revenue}_T = 1,000,000 \times S_{\mathcal{F}/\text{USD},T}$$

- **Pros:** Full upside benefit if $\mathcal{F}$ appreciates ($S_T \uparrow$).
- **Cons:** Severe downside risk if $\mathcal{F}$ depreciates ($S_T \downarrow$).

2. Forward Contract Hedge

Payoff at T (in USD):

$$\text{Revenue}_T = 1,000,000 \times F_{\mathcal{F}/\text{USD},0,T}$$

- **Pros:** For a certain amount/date, the contractual conversion rate is locked; an at-market forward generally requires no premium.
- **Cons:** Forfeits all upside gain if $\mathcal{F}$ appreciates.

A Money Market Hedge creates a synthetic forward by matching money market borrowing/lending with spot FX transactions.

MMH for Foreign Receivable ($\mathcal{F}$)

Goal: Receive H currency today from future $\mathcal{F}$ revenue.

- 1 Borrow $\mathcal{F}$ today: $\frac{\text{Receivable}}{1+i_{\mathcal{F}} \cdot t}$.
- 2 Convert borrowed $\mathcal{F}$ to H at spot rate $S_{\mathcal{F}/H,0}$.
- 3 Invest H at domestic interest rate i_H .
- 4 At date T : Repay $\mathcal{F}$ loan using the incoming receivable.

MMH for Foreign Payable ($\mathcal{F}$)

Goal: Pay exact $\mathcal{F}$ liability at date T .

- 1 Calculate required $\mathcal{F}$ deposit today: $\frac{\text{Payable}}{1+i_{\mathcal{F}} \cdot t}$.
- 2 Convert H to $\mathcal{F}$ at spot rate $S_{\mathcal{F}/H,0}$.
- 3 Borrow H today to fund the spot conversion.
- 4 At date T : Use $\mathcal{F}$ deposit proceeds to settle payable.

Under CIP, $F_{\mathcal{F}/H} = S_{\mathcal{F}/H} \cdot \frac{1+i_H t}{1+i_{\mathcal{F}} t}$. The payoffs match at the same borrowing and lending rates. Spreads, credit, collateral, taxes, and funding constraints can create differences.

Algebraic CIP Proof →

Currency options provide **asymmetric risk protection** by offering the right, but not the obligation, to trade at exercise price K .

Exposure Type	Option Choice	Effective Payoff per Unit	Net Effect
Receivable ($\mathcal{F}$)	Put Option (Strike K , Premium p)	$\max(S_T, K) - p(1 + i_H t)$	Floor revenue at $K - p(1 + i_H t)$ while retaining upside if $S_T > K$.
Payable ($\mathcal{F}$)	Call Option (Strike K , Premium p)	$\min(S_T, K) + p(1 + i_H t)$	Cap cost at $K + p(1 + i_H t)$ while retaining benefit if $S_T < K$.

- **Upfront Premium:** Unlike forwards, options require paying an immediate cash premium p .
- **Opportunity Cost of Premium:** $p(1 + i_H t)$ represents the future value of the upfront premium at date T .

Option payoff illustration →

Option Pricing & Greeks Reference →

An FX swap combines an exchange of two currencies on a **near date** with a reverse exchange on a **far date**. Both rates are agreed when the trade is booked.

- **Spot/forward swap**: exchange currencies at spot, then reverse the exchange at the contracted forward rate.
- **Forward/forward swap**: both exchanges take place in the future. Choose the dates to match the firm's cash flows.
- **Funding use**: obtain one currency temporarily by delivering another, with an obligation to reverse the exchange.
- A standard swap usually keeps the amount in one currency constant. The amount in the other changes with the near and far rates.

Example direction: a firm that needs EUR now and will receive EUR later buys EUR on the near date and sells EUR on the far date.

Source: BIS glossary: FX swaps. Here $t = 0$ denotes the near settlement date.

FX Swap: Matching a Payable and a Later Receipt

A US firm must pay **EUR 1M on day 10** and expects to receive **EUR 1M on day 20**. It books a forward/forward swap today.

Cash flow from the firm's perspective	Day 10	Day 20
Commercial EUR cash flow	-1,000,000	+1,000,000
EUR cash flow from swap	+1,000,000	-1,000,000
Net EUR cash flow	0	0
USD cash flow from swap	-1,100,000	+1,101,000

- Illustrative agreed rates: $K_{10} = 1.1000$ and $K_{20} = 1.1010$ USD per EUR. The firm must fund the USD payment on day 10.
- The USD 1,000 difference is across *different dates*. Compare it with the firm's USD funding cost before calling it a gain.
- If the export receipt arrives late, the firm still owes EUR on day 20. Matching the amount alone does not remove timing risk.

Use S_0 and $F_{0,T}$ in **USD per EUR**. For simple annual rates and maturity $\tau = T$ in years, frictionless CIP gives

$$F_{0,T} = S_0 \frac{1 + i_{\text{USD}}\tau}{1 + i_{\text{EUR}}\tau}.$$

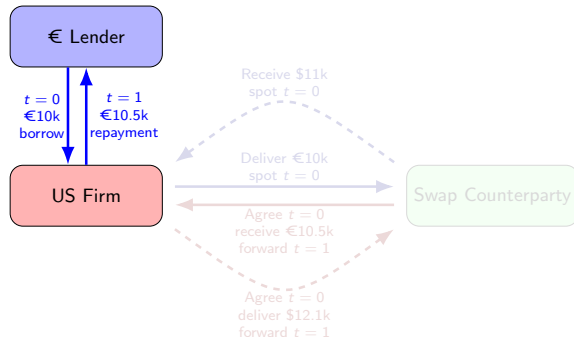
- With $S_0 = 1.10$, $i_{\text{EUR}} = 5\%$, $i_{\text{USD}} = 10\%$, and $\tau = T = 1$:

$$F_{0,1} = 1.10 \frac{1.10}{1.05} = 1.152381 \text{ USD per EUR.}$$

- **Swap points** are $F - S = 0.052381$ USD per EUR. Higher USD interest raises the USD-per-EUR forward rate under these assumptions.
- The forward rate prices a covered exchange. It does not by itself predict future spot.
- A firm's own borrowing spread affects its funding cost. Market forward pricing uses the relevant market funding rates, with any basis and dealing costs.

Source: BIS, Covered interest parity lost (2016).

One-year loan, zero borrowing spreads, and CIP. The US firm borrows EUR 10,000 but needs USD.



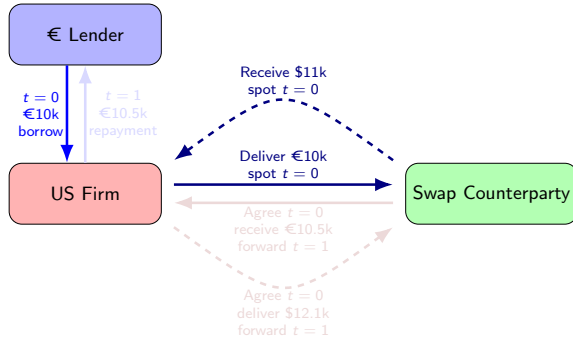
- Borrow at 5% in EUR. The USD rate is 10%.
- At $t = 0$, exchange EUR 10,000 for USD 11,000 at $S_0 = 1.10$.
- At $t = 1$, buy the full EUR 10,500 loan repayment at the agreed forward rate:

$$10,500 \times \underbrace{1.10 \frac{1.10}{1.05}}_{F_{0,1}} = \$12,100.$$

- Net result: receive USD 11,000 now and repay USD 12,100 in one year, a 10% USD funding cost.

Principal and interest hedge: the far EUR amount is larger than the near amount. A standard EUR 10,000 FX swap plus a forward purchase of EUR 500 gives these same cash flows.

One-year loan, zero borrowing spreads, and CIP. The US firm borrows EUR 10,000 but needs USD.



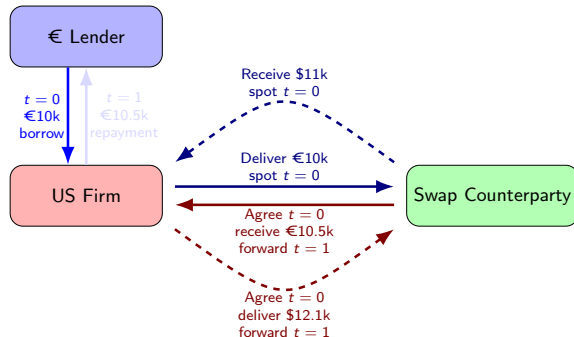
- Borrow at 5% in EUR. The USD rate is 10%.
- At $t = 0$, exchange EUR 10,000 for USD 11,000 at $S_0 = 1.10$.
- At $t = 1$, buy the full EUR 10,500 loan repayment at the agreed forward rate:

$$10,500 \times \underbrace{1.10 \frac{1.10}{1.05}}_{F_{0,1}} = \$12,100.$$

- Net result: receive USD 11,000 now and repay USD 12,100 in one year, a 10% USD funding cost.

Principal and interest hedge: the far EUR amount is larger than the near amount. A standard EUR 10,000 FX swap plus a forward purchase of EUR 500 gives these same cash flows.

One-year loan, zero borrowing spreads, and CIP. The US firm borrows EUR 10,000 but needs USD.



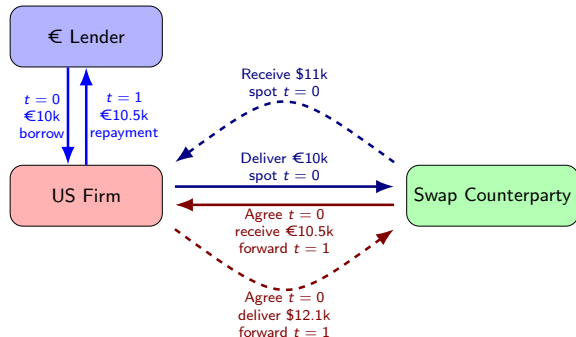
- Borrow at 5% in EUR. The USD rate is 10%.
- At $t = 0$, exchange EUR 10,000 for USD 11,000 at $S_0 = 1.10$.
- At $t = 1$, buy the full EUR 10,500 loan repayment at the agreed forward rate:

$$10,500 \times \underbrace{1.10 \frac{1.10}{1.05}}_{F_{0,1}} = \$12,100.$$

- Net result: receive USD 11,000 now and repay USD 12,100 in one year, a 10% USD funding cost.

Principal and interest hedge: the far EUR amount is larger than the near amount. A standard EUR 10,000 FX swap plus a forward purchase of EUR 500 gives these same cash flows.

One-year loan, zero borrowing spreads, and CIP. The US firm borrows EUR 10,000 but needs USD.



- Borrow at 5% in EUR. The USD rate is 10%.
- At $t = 0$, exchange EUR 10,000 for USD 11,000 at $S_0 = 1.10$.
- At $t = 1$, buy the full EUR 10,500 loan repayment at the agreed forward rate:

$$10,500 \times \underbrace{1.10 \frac{1.10}{1.05}}_{F_{0,1}} = \$12,100.$$

- Net result: receive USD 11,000 now and repay USD 12,100 in one year, a 10% USD funding cost.

Principal and interest hedge: the far EUR amount is larger than the near amount. A standard EUR 10,000 FX swap plus a forward purchase of EUR 500 gives these same cash flows.

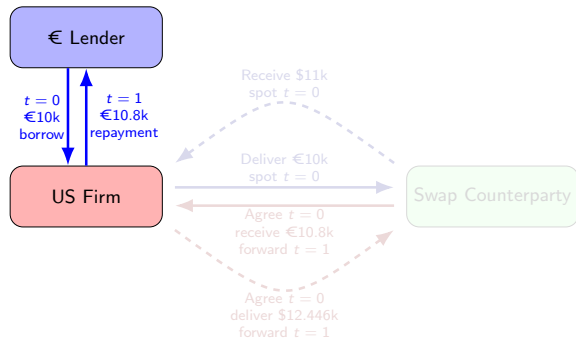
The EUR Cash Flows Cancel at Each Date

All amounts below are cash flows to the US firm. Positive means a receipt. The loan and the hedge must be evaluated together.

Transaction	Today ($t = 0$)		One year ($t = 1$)	
	EUR	USD	EUR	USD
EUR loan	+10,000	0	-10,500	0
Spot exchange	-10,000	+11,000	0	0
Forward exchange	0	0	+10,500	-12,100
Net cash flow	0	+11,000	0	-12,100

- If the firm hedged only EUR 10,000 at maturity, it would still need to buy **EUR 500 of interest** at the future spot rate.
- With the full repayment hedged, even a large change in S_1 leaves the contracted USD repayment unchanged, assuming the counterparty performs.

Same market rates: 5% in EUR and 10% in USD. The firm pays an extra 3 pct pts in EUR or 6 in USD.



- Borrow EUR 10,000 at 8% and exchange it for USD 11,000.
- Buy EUR 10,800 forward to repay the EUR loan. CIP still uses the market rates:

$$10,800 \times \left(1.10 \frac{1.10}{1.05} \right) \approx \$12,445.71.$$

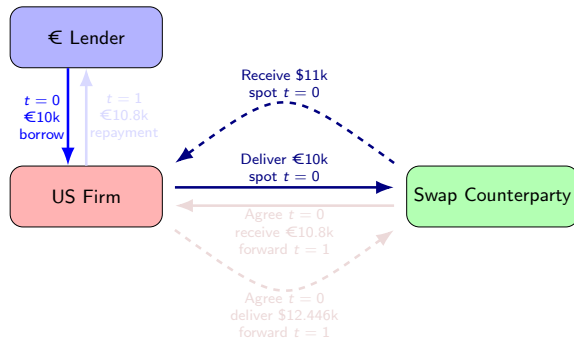
- Synthetic USD borrowing rate:

$$\frac{12,445.71}{11,000} - 1 \approx 13.143\%.$$

- Direct USD repayment at 16%: USD 12,760. The hedged EUR route saves **USD 314.29** before fees.

Forward purchase: EUR 10,000 principal plus EUR 800 interest. Ignore dealing fees and collateral costs.

Same market rates: 5% in EUR and 10% in USD. The firm pays an extra 3 pct pts in EUR or 6 in USD.



- Borrow EUR 10,000 at 8% and exchange it for USD 11,000.
- Buy EUR 10,800 forward to repay the EUR loan. CIP still uses the market rates:

$$10,800 \times \left(1.10 \frac{1.10}{1.05} \right) \approx \$12,445.71.$$

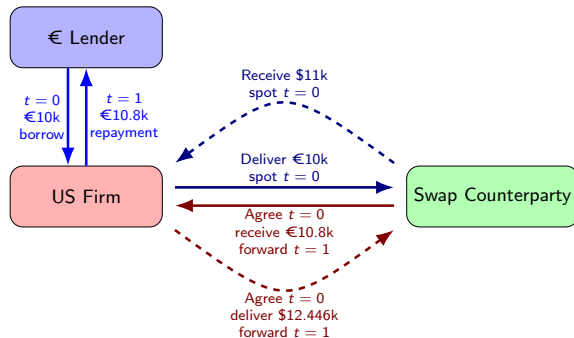
- Synthetic USD borrowing rate:

$$\frac{12,445.71}{11,000} - 1 \approx 13.143\%.$$

- Direct USD repayment at 16%: USD 12,760. The hedged EUR route saves **USD 314.29** before fees.

Forward purchase: EUR 10,000 principal plus EUR 800 interest. Ignore dealing fees and collateral costs.

Same market rates: 5% in EUR and 10% in USD. The firm pays an extra 3 pct pts in EUR or 6 in USD.



- Borrow EUR 10,000 at 8% and exchange it for USD 11,000.
- Buy EUR 10,800 forward to repay the EUR loan. CIP still uses the market rates:

$$10,800 \times \left(1.10 \frac{1.10}{1.05} \right) \approx \$12,445.71.$$

- Synthetic USD borrowing rate:

$$\frac{12,445.71}{11,000} - 1 \approx 13.143\%.$$

- Direct USD repayment at 16%: USD 12,760. The hedged EUR route saves **USD 314.29** before fees.

Forward purchase: EUR 10,000 principal plus EUR 800 interest. Ignore dealing fees and collateral costs.

Hedged Borrowing: Separate the Loan Rate from the Forward

- ① **Loan rate:** EUR market rate 5% plus the firm's spread 3 percentage points gives 8%. EUR 10,000 grows to $10,000(1.08) = 10,800$ owed.
- ② **Forward pricing:** market rates remain 5% EUR and 10% USD. Thus $F = 1.10(1.10)/1.05 = 1.15238095 \dots$ USD per EUR.
- ③ **Cover the entire debt:** buy EUR 10,800 forward. Locked USD repayment: $10,800F = 12,445.71$.
- ④ **Measure the USD rate:** USD proceeds today are $10,000(1.10) = 11,000$. The cost is $12,445.71/11,000 - 1 = 13.143\%$.
- ⑤ **Compare like dates and amounts:** direct USD borrowing costs $10\% + 6\% = 16\%$, repaying $11,000(1.16) = 12,760$. Savings: USD 314.29.

Forwards, FX Swaps, and Cross-Currency Swaps

Instrument	Contracted exchanges	Typical corporate use
Outright forward	One currency exchange at a future date	Lock the home-currency value of a known foreign payable or receivable.
FX swap	Near exchange and reverse far exchange, usually with one currency amount held fixed	Bridge a currency funding gap between two dates.
Cross-currency interest-rate swap	Currency principal exchanges plus periodic interest payments	Convert a stream of foreign bond or loan payments into the desired currency.

The loan diagrams combine a spot exchange with a forward covering **principal plus interest**. With several coupon dates, each payment needs cover.

Source: BIS, FX swaps and forwards: missing global debt? (2017).

FX Swap Risks and the Funding Decision

- **Amount and date risk:** a late or smaller foreign receipt leaves the firm short of the currency it promised to deliver.
- **Rollover risk:** a three-month swap against a five-year asset fixes only the first three months. Renewing the hedge may become costly or unavailable.
- **Counterparty and liquidity risk:** a counterparty can fail, and collateral calls can require cash before the underlying asset pays out.
- **All-in cost:** compare the hedged foreign loan with direct borrowing after spreads, swap pricing, fees, and collateral funding costs.

Foreshadowing Week 8: banks also use FX swaps to fund foreign-currency assets. We will study how funding demand, dealer capacity, and the cross-currency basis affect their cost.

FX Swap Check: Hedging the Whole Loan

A US firm needs USD 11,000 for one year. It can borrow EUR 10,000 at 8% and exchange the proceeds at $S_0 = 1.10$ USD per EUR. Its forward quote is $F_{0,1} = 1.15$ USD per EUR.

- ① How many EUR should it buy forward, and why?
- ② What is the locked USD repayment and the synthetic USD borrowing rate?
- ③ Direct USD borrowing costs 14%. Which route is cheaper before fees?
- ④ If it buys only EUR 10,000 forward and $S_1 = 1.30$, what is its total USD repayment?

Use the quoted forward directly. Assume a single payment at maturity and no fees. This is a separate numerical exercise from the preceding CIP example.

Worked Solution →

EUR Receivable: Unhedged and Forward Cash Flows

A Hong Kong firm receives EUR 1M in one year. The quote is HKD per EUR. Today $S_0 = 8.50$ and the one-year forward is $F_0 = 8.60$.

- ① **Identify the risk:** EUR depreciation reduces the HKD value of the receipt.
- ② **Unhedged:** if $S_T = 7.80$, convert EUR 1M for $1,000,000 \times 7.80 = 7,800,000$ HKD.
- ③ **Forward today:** agree to *sell* EUR 1M in one year for $1,000,000 \times 8.60 = 8,600,000$ HKD.
- ④ **Forward at maturity:** receive EUR 1M from the customer, deliver it to the dealer, and receive HKD 8.6M. The EUR inflow and outflow cancel.
- ⑤ At $S_T = 9.20$, the unhedged receipt would be HKD 9.2M. The forward still pays HKD 8.6M.

EUR Receivable: Money Market Hedge Today

Same EUR 1M receipt in one year. EUR borrowing rate 3.77%, HKD deposit rate 5%, spot 8.50 HKD per EUR. Ignore spreads and fees.

- ① **Work backward from EUR 1M:** borrow $1,000,000 / 1.0377 = 963,669.65$ EUR today. This loan will grow to exactly EUR 1M.
- ② **Sell those EUR today:** $963,669.65 \times 8.50 = 8,191,192.06$ HKD. Multiplication converts EUR into HKD.
- ③ **Deposit the HKD:** $8,191,192.06 \times 1.05 = 8,600,751.66$ HKD in one year.
- ④ Nothing depends on the future spot rate. We have fixed both the EUR loan repayment and the HKD deposit proceeds.

Calculations retain full precision before rounding the displayed amounts.

EUR Receivable: Money Market Hedge at Maturity

- ① **Customer pays:** receive EUR 1,000,000.
- ② **Repay the EUR loan:** $963,669.654 \dots \times 1.0377 = 1,000,000$ EUR. Net EUR cash flow is zero.
- ③ **Collect the HKD deposit:** receive HKD 8,600,751.66. This is the locked home-currency receipt.
- ④ **Compare with the forward:** HKD 8,600,751.66 versus HKD 8,600,000. The small difference comes from the rounded interest-rate inputs.
- ⑤ Exact CIP at these rates gives $F = 8.50(1.05)/1.0377 = 8.60075166 \dots$. Using that rate, the two methods match exactly.

EUR Receivable: Put Premium and a Weak Euro

Same EUR 1M receipt. Buy EUR puts with strike 8.50 and upfront premium 0.20 HKD per EUR. Compare all amounts in one year, using 5% HKD interest.

- ① **Premium today:** $1,000,000 \times 0.20 = 200,000$ HKD.
- ② **Premium at maturity:** $200,000 \times 1.05 = 210,000$ HKD. This includes the premium's financing or opportunity cost.
- ③ If $S_T = 7.80$, exercise the right to sell at 8.50. Gross proceeds are $1,000,000 \times 8.50 = 8,500,000$ HKD.
- ④ **Net receipt:** $8,500,000 - 210,000 = 8,290,000$ HKD.
- ⑤ Check another way: spot receipt HKD 7.8M plus put payoff $1M(8.50 - 7.80) = 0.7M$, less premium FV 0.21M, also gives 8.29M.

EUR Receivable: Put Upside and the Forward Break-Even

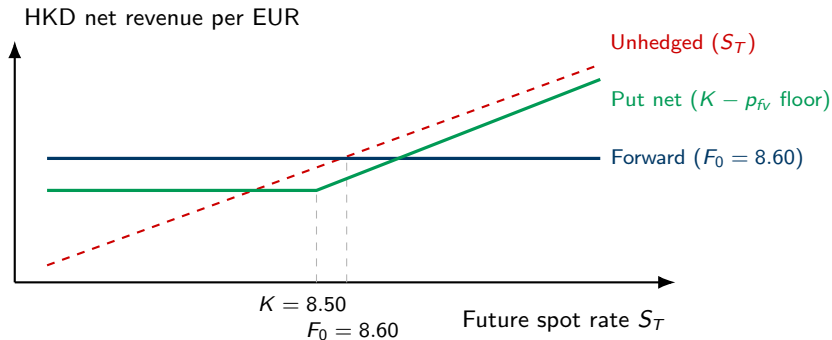
- ① If $S_T = 9.20 > 8.50$, let the put expire and sell EUR 1M at spot: $1,000,000 \times 9.20 = 9,200,000$ HKD.
- ② The premium is paid even if the option expires. Net receipt: $9,200,000 - 210,000 = 8,990,000$ HKD.
- ③ The forward pays HKD 8.6M. For a spot rate above the strike, set the two net receipts equal:
$$1,000,000S_T - 210,000 = 8,600,000.$$
- ④ Add 210,000 to both sides, then divide by 1,000,000: $S_T = 8.81$ HKD per EUR.
- ⑤ The put hedge beats the forward only above 8.81. Between 8.50 and 8.81, it expires but its net receipt is still lower.

Setup: A HK firm expects a **€1,000,000** receivable in 1 year ($T = 1$).

- Spot rate $S_0 = 8.50$ HKD per EUR. 1-year Forward rate $F_0 = 8.60$ HKD per EUR.
- 1-year Put Option: Strike $K = 8.50$ HKD per EUR, Premium $p = 0.20$ HKD per EUR (Future value of premium ≈ 0.21 HKD per EUR).
- Interest rates: $i_{\text{HKD}} = 5\%$, $i_{\text{EUR}} = 3.77\%$.

Future Spot Rate S_T	Unhedged Net Revenue	Forward Net Revenue	Put Option Action	Put Net Revenue (less Prem FV)
$S_T = 7.80$ (EUR Depreciates)	7,800,000 HKD	8,600,000 HKD	Exercise at 8.50	8,290,000 HKD
$S_T = 8.50$ (Unchanged)	8,500,000 HKD	8,600,000 HKD	Indifferent at strike	8,290,000 HKD
$S_T = 8.80$ (EUR Appreciates)	8,800,000 HKD	8,600,000 HKD	Expire (Sell at 8.80)	8,590,000 HKD
$S_T = 9.20$ (EUR Rallies)	9,200,000 HKD	8,600,000 HKD	Expire (Sell at 9.20)	8,990,000 HKD

Full Payoff Comparison Matrix →



Below Strike ($S_T < 8.50$): Forward pays 8.60. Put net receipts are 8.29 and exceed unhedged receipts only when $S_T < 8.29$.

Break-Even Point: Put option outperforms forward only if $S_T > F_0 + p_{fv} = 8.60 + 0.21 = 8.81$ HKD per EUR.

A payable hedge from the funding legs

Coleman owes EUR 100,000 in one year. EUR/USD spot is 1.18, the forward is 1.20, the firm's USD borrowing rate is 8%, and its EUR deposit rate is 5%.

Step	Amount
Deposit EUR today	$100,000/1.05 = 95,238.10$
Borrow USD to buy the deposit	$95,238.10(1.18) = 112,380.95$
Repay USD in one year	$112,380.95(1.08) = 121,371.43$
Alternative forward payment	$100,000(1.20) = 120,000$

- The forward costs USD 1,371.43 less at maturity. These firm-specific rates need not satisfy frictionless CIP at the quoted forward.
- An arbitrage conclusion would also require executable borrowing and lending quotes in the reverse direction, credit access, and transaction costs.

EUR Payable: Why We Deposit EUR Today

- ① **Start from the bill:** EUR 100,000 is due in one year. We need an asset that pays exactly that amount on that date.
- ② **Discount at the EUR deposit rate:** $100,000/1.05 = 95,238.10$ EUR today. Check:
 $95,238.1 \times 1.05 = 100,000$.
- ③ **Buy the deposit with borrowed USD:** $95,238.1 \times 1.18 = 112,380.95$ USD today.
- ④ **At maturity:** the EUR deposit pays the supplier. Separately, repay the USD loan:
 $112,380.952 \dots \times 1.08 = 121,371.43$ USD.
- ⑤ Buying EUR 100,000 forward at 1.20 costs USD 120,000 on the same date. Saving:
 $121,371.43 - 120,000 = 1,371.43$ USD.

Call options on a payable

Continue with EUR 100,000 owed. A EUR call has strike 1.20 USD per EUR and an upfront premium of 0.03 USD per EUR. Finance the premium at 8% for one year.

$$\text{USD cost at maturity} = 100,000 \min(S_T, 1.20) + 3,000(1.08).$$

Scenario	Probability	Cost
EUR/USD = 1.16	20%	USD 119,240
EUR/USD = 1.24	80%	USD 123,240

- Expected cost is USD 122,440 under these stated probabilities. The option caps the cost at USD 123,240 and retains the benefit of a cheaper euro.
- The USD 120,000 forward is cheaper in expectation in this example. Different probabilities or an uncertain underlying payment can change the choice.

EUR Call: Calculate Each Scenario Before Taking the Average

EUR 100,000 payable, strike 1.20 USD per EUR, premium 0.03, and 8% premium financing rate.

- ① Premium today: $100,000 \times 0.03 = 3,000$ USD. Repayment in one year: $3,000 \times 1.08 = 3,240$ USD.
- ② At $S_T = 1.16$, buy EUR at spot and let the call expire. Total cost:
 $100,000 \times 1.16 + 3,240 = 119,240$ USD.
- ③ At $S_T = 1.24$, exercise the call to buy EUR at 1.20. Total cost: $100,000 \times 1.20 + 3,240 = 123,240$ USD.
- ④ Weight the *total costs*: $0.20(119,240) + 0.80(123,240) = 23,848 + 98,592 = 122,440$ USD.
- ⑤ The USD 123,240 cap includes the premium. USD 120,000 is only the cost of exercising the call.

Put options on a receivable

A firm will receive AUD 200,000. Buy a put with strike 0.72 USD per AUD and premium 0.02 USD per AUD. For this exercise only, ignore interest on the premium.

AUD/USD at maturity	Net USD per AUD	Net USD
0.71	0.70	140,000
0.74	0.72	144,000
0.76	0.74	148,000

- With equal probabilities of one third, expected receipts are USD 144,000. Without probabilities, an expected value cannot be calculated.
- A forward locking USD 142,000 gives certainty but no upside. A higher expected option receipt does not by itself establish a better risk-adjusted choice.
- With a positive financing rate, subtract the future value of the premium to compare all cash flows at maturity.

Farmer hedge example →

AUD Put: Build the Three Net Receipts

AUD 200,000 receivable, strike 0.72 USD per AUD, premium 0.02. Ignore interest on the premium in this exercise.

- ① Premium: $200,000 \times 0.02 = 4,000$ USD.
- ② At 0.71, exercise: $200,000 \times 0.72 - 4,000 = 140,000$ USD.
- ③ At 0.74, let the put expire: $200,000 \times 0.74 - 4,000 = 144,000$ USD.
- ④ At 0.76, let it expire: $200,000 \times 0.76 - 4,000 = 148,000$ USD.
- ⑤ With equal probabilities, $(140,000 + 144,000 + 148,000)/3 = 144,000$ USD. This is an average across scenarios, not a guaranteed receipt.

Game Round 4 Rules & Procedure

Scenario: You are the CFO of a Hong Kong trading firm expecting a **€1,000,000 receivable** in 90 days.

- Spot rate $S_0 = 8.40$ HKD per EUR. 90-day Forward $F_0 = 8.42$ HKD per EUR.
- 90-day Put Option: Strike $K = 8.40$, Premium $p = 0.12$ HKD per EUR ($p_{fV} = 0.1215$).



[Open Week 4 Game](#)

Game Decision Stages:

- 1 **Stage 1:** [Open the game](#) and log CUHK ID. Choose strategy: *Unhedged, Forward, Money Market Hedge, or Put Option*.
- 2 **Stage 2:** Compute locked net cash flows vs option floor.
- 3 **Stage 3 [Exogenous Shock]:** European Central Bank unexpectedly cuts rates; EUR depreciates to $S_T = 7.80$ HKD per EUR!
- 4 **Stage 4 & 5:** Payoff debrief & submit 1-sentence Exit Reflection:
“Why did you prefer a hard forward lock over option flexibility for this exposure?”

How should a firm choose between Forward, Option, or Unhedged positions?

Strategy	Optimal Conditions	Corporate Considerations
Forward Hedge	Certain, committed cash flow; high cost of cash-flow volatility.	No option premium; fixes the contractual rate but retains counterparty/basis risks.
Option Hedge	Contingent/uncertain cash flow (e.g. bidding on a contract); high upside potential.	Requires upfront cash budget; costly premium acts as hurdle.
Unhedged	Exposure is immaterial, naturally offset, or within an approved risk tolerance.	Avoids hedge costs but retains cash-flow volatility; a directional FX forecast alone is not a risk-management rationale.

Corporate Hedging Policies & Accounting →

Hedging is not a free lunch and comes with key operational limits:

① Quantity Uncertainty (Overhedging):

- If expected sales are €10M but actual sales are only €6M, a €10M forward hedge leaves €4M exposed to speculative FX risk!

② Short-Horizon Limitations:

- Hedging a 3-month transaction protects current margin, but repeated short-term hedges cannot protect a firm against persistent 5-year currency trends.

③ Operational / Natural Hedging:

- Possible complement: Match foreign revenue with foreign expenses (e.g. source inputs or borrow in the revenue currency), after weighing operational and financing costs.

Natural Hedging & Restructuring Details →

Overhedging: Find the Missing EUR Before Calculating Profit

Illustrative one-year hedge: sell EUR 10M forward at 8.60 HKD per EUR. Actual customer receipts are only EUR 6M.

- ① The forward still requires delivery of EUR 10M. Shortfall: $10 - 6 = 4$ million EUR.
- ② At $S_T = 9.20$, buy the missing EUR 4M for $4 \times 9.20 = 36.8$ million HKD.
- ③ Deliver EUR 10M into the forward and receive $10 \times 8.60 = 86$ million HKD. Net cash receipt: $86 - 36.8 = 49.2$ million HKD.
- ④ Hedging only the actual EUR 6M would have paid $6 \times 8.60 = 51.6$ million HKD. The extra forward position loses $51.6 - 49.2 = 2.4$ million HKD.
- ⑤ Check: $4M(8.60 - 9.20) = -2.4M$ HKD. A hedge can create exposure when the underlying amount changes.

Economic exposure measures how real exchange rate changes affect a firm's future operational cash flows, pricing power, and long-term market value.

Key Insights

- **Broad Scope:** Affects future uncommitted transactions and market competitiveness.
- **Domestic Impact:** Even a firm with zero foreign sales faces economic exposure if foreign competitors become cheaper when their home currency weakens!

Example: Retailer Competitiveness

A US retailer prices goods in USD world-wide.

When the EUR depreciates against USD, European consumers face higher local prices for USD goods, reducing sales volume even though transaction prices were in USD.

Extended Case Study: Madison Co. →

Operating exposure across business models

- A foreign-currency export receivable loses home-currency value when that foreign currency depreciates. A foreign-currency import payable becomes cheaper.
- An exporter invoicing in home currency can still lose demand when foreign buyers face higher local prices.
- A domestic-only producer can face cheaper imported competition. An importer may benefit from cheaper inputs but also face price competition.
- The net effect depends on invoicing, pass-through, demand elasticities, competitors, and the currencies of revenues, costs, and debt.

Economic Exposure Without a EUR Invoice

Illustration: a US exporter keeps its price at USD 120. Initially EUR/USD is 1.20 USD per EUR. Later it falls to 1.00.

- ① Initially, the European customer pays $120/1.20 = 100$ EUR. Divide because we are converting USD into EUR.
- ② After EUR depreciation, the same USD 120 price costs $120/1.00 = 120$ EUR, a 20% increase for the customer.
- ③ If the exporter keeps the USD price, customers may buy less. The firm has demand risk even though it invoices in USD.
- ④ Alternatively, keeping the customer price at EUR 100 means collecting $100 \times 1.00 = 100$ USD, down from USD 120.

Translation exposure arises when converting financial statements of foreign subsidiaries into parent company home currency for consolidated reporting.

- **Mechanism:** Assets, liabilities, and earnings of foreign subsidiaries are translated at closing or average exchange rates.
- **Accounting Impact:** Under the current-rate translation method, translation adjustments generally enter OCI/equity; remeasurement under the temporal method can affect earnings.
- **Hedging Choice:** A net-investment hedge can reduce translation volatility and may receive hedge-accounting treatment, but it introduces funding, basis, liquidity, and counterparty risks. The objective should be explicit.

ASC 830 Translation & Remeasurement Reference →

Translation exposure and a share-price illustration

Providence has a UK subsidiary earning GBP 10M in each year. Suppose the average GBP/USD rate falls from 1.70 to 1.50.

Item	Year 1	Year 2
Translated earnings	USD 17M	USD 15M
EPS, assuming 10M shares	USD 1.70	USD 1.50
Price, assuming a constant P/E of 20	USD 34	USD 30

- The USD 4 price change follows the assumed constant P/E. Market prices need not mechanically follow translated accounting earnings.
- Translation of a foreign functional currency generally sends the currency translation adjustment to other comprehensive income. Remeasurement rules differ.
- Hedging forecast earnings with a forward creates real cash gains or losses and may not match average-rate accounting or qualify as a net-investment hedge.

Translation Illustration: Earnings, EPS, and Assumed Price

- ① **Translate earnings:** GBP 10M at 1.70 USD per GBP gives USD 17M. At 1.50, it gives USD 15M. GBP earnings did not change.
- ② **Divide by shares:** USD 17M / 10M shares = USD 1.70 per share. USD 15M / 10M shares = USD 1.50 per share.
- ③ **Apply the stated P/E assumption:** $1.70 \times 20 = 34$ USD and $1.50 \times 20 = 30$ USD per share.
- ④ **Compare:** assumed price falls by $34 - 30 = 4$ USD. The percentage is $4/34 = 11.76\%$.
- ⑤ The price result assumes the P/E stays at 20. Translation alone does not establish an actual market-price change or a cash loss of the same amount.

Weeks 1–2: Quotes and Parity

Week 1: Executable FX trades

- Read quote units, reciprocal changes, and executable cross rates. Trace locational and triangular arbitrage.
- Execute market and limit orders using bid/ask, depth, and slippage. Explain dealer inventory and trade-size limits.

Week 2: Pricing and expectations

- CIP: forward points, day counts, arbitrage in both directions, funding spreads, depth, and synthetic funding.
- UIP: forwards versus forecasts, scenario returns and downside risk, policy surprises, and Fama evidence.
- Goods prices: LOP, absolute/relative PPP, real exchange rates, Balassa–Samuelson, and assumptions behind real interest parity.

Week 3: Interpreting FX Markets

- Explain policy-path surprises and overshooting. Calculate carry returns and stress leverage, funding appreciation, and liquidity needs.
- Compare carry, momentum, value, and macro signals. Separate forecast accuracy from net returns. Evaluate diversification, order flow, positioning, and tests using only information available then.
- Read option payoffs, implied volatility, term structure, smile/skew, and risk reversals. Match protection to the currency and maturity.
- Apply the trilemma and HKMA mechanics. Distinguish intervention, sterilization, and policy signals. Assess outcomes against the stated objective.

Scope: Main-slide methods and interpretations. Appendix derivations and historical details are reference material. Week 4 review follows.

Hedge Calculations

- 1 Calculate net payoffs for Unhedged, Forward, Money Market Hedge (MMH), and Options.
- 2 Identify direction of exposure (Receivable vs Payable) and select correct option type (Put vs Call).
- 3 Explain the idealized MMH–forward equivalence under Covered Interest Parity (CIP).
- 4 Derive break-even exchange rates between Forward and Option hedges.
- 5 Trace both FX swap legs, hedge principal plus interest, and compare synthetic with direct borrowing costs.

Exposure and Risk

- Net cash flows by currency and date. Use signed exposures and common-currency values for portfolio risk.
- Interpret correlation, VaR, and stress tests. A loss quantile is not a maximum loss.
- Distinguish transaction, operating, and translation exposure. Explain overhedging, natural hedges, and residual funding or collateral risks.

Next Week (Week 5): International Capital Budgeting & Foreign Project Valuation!

Appendix: Currency Return Calculations

- ① **EUR, date 0 to 1:** divide the new price by the old price, then subtract 1:
 $1.1110/1.1000 - 1 = 1.01 - 1 = 0.01 = 1\%$.
- ② **GBP, date 0 to 1:** $1.2750/1.2500 - 1 = 1.02 - 1 = 0.02 = 2\%$. Plot the point (1, 2) in percentage points.
- ③ **EUR, date 1 to 2:** $1.09989/1.1110 - 1 = 0.99 - 1 = -1\%$. Use the *previous date*, not date 0.
- ④ **GBP, date 1 to 2:** $1.26225/1.2750 - 1 = -1\%$. Plot (-1, -1). Both currencies fell against USD on this date.

← Back to Main Slide

Appendix: Correlation and Sample Moments

Use percentage-point returns $X = (1, -1, 2)$ and $Y = (2, -1, 3)$. There are $n = 3$ observations.

① Means: $\bar{X} = (1 - 1 + 2)/3 = 2/3$, $\bar{Y} = (2 - 1 + 3)/3 = 4/3$.

② Deviations: $X - \bar{X} = (1/3, -5/3, 4/3)$ and $Y - \bar{Y} = (2/3, -7/3, 5/3)$.

③ Divide squared deviations by $n - 1 = 2$:

$$s_X^2 = \frac{(1 + 25 + 16)/9}{2} = \frac{7}{3}, \quad s_Y^2 = \frac{(4 + 49 + 25)/9}{2} = \frac{13}{3}.$$

④ Covariance: $s_{XY} = [(2 + 35 + 20)/9]/2 = 19/6$.

⑤ $s_X = 1.528$, $s_Y = 2.082$, so $\rho = (19/6)/(1.528 \times 2.082) \approx 0.996$.

The currencies move closely together, but their standard deviations differ.

← Back to Main Slide

Appendix: FX Swap Check Solution

- ① **Full EUR repayment:** $10,000 \times 1.08 = \text{EUR } 10,800$. The forward must cover the interest as well as the principal.
- ② **Locked USD repayment:** $10,800 \times 1.15 = \$12,420$. The effective one-year USD borrowing rate is

$$\frac{12,420}{11,000} - 1 = 12.909\%.$$

- ③ **Direct USD loan:** $11,000 \times 1.14 = \$12,540$. Hedged EUR borrowing saves **USD 120** before fees and collateral costs.
- ④ **Principal-only hedge:** pay $10,000 \times 1.15 = \$11,500$ under the forward, then buy EUR 800 at spot for $800 \times 1.30 = \$1,040$. Total repayment is **USD 12,540**. The unhedged interest changes the outcome.

← Back to Main Slide

Appendix: Comparing the Three Types of FX Exposure

[Background Context]

Dimension	Transaction Exposure	Economic Exposure	Translation Exposure
Definition	Risk on contracted, committed future cash flows.	Risk on future operational cash flows and market value.	Risk on consolidated accounting financial statements.
Time Horizon	Until contracted cash flow settles (short or long dated).	Long-term (continuous operational horizon).	Periodic reporting dates (Quarterly / Annual).
Primary Impact	Real cash flows and realized profit margins.	Real operational cash flows and competitive position.	Reported equity/OCI and, under some remeasurement methods, earnings.
Primary Tools	Forwards, MMH, Options, Collars.	Operational restructuring, currency matching.	Disclosure, matching foreign debt assets.

← Back to Main Slide

Appendix: Multilateral Netting & Portfolio Example

Suppose a HK multinational has subsidiaries in the US, Europe, and UK with the following scheduled 30-day cash flows:

- US Parent owes €2M to German sub; expects €5M from French sub. Net EUR = +€3M (**Long**).
- US Parent owes £4M to UK sub; expects £1M from UK customer. Net GBP = -£3M (**Short**).

Analytical Steps:

- ① **Single-Currency Netting:** Internal settlement reduces EUR transactions from €7M total to €3M net. Saves bank bid-ask spreads on €4M!
- ② **Cross-Currency Correlation:** With positively correlated EUR/USD and GBP/USD changes, opposite-signed EUR and GBP exposures can reduce portfolio variance. The degree of offset depends on home-currency notionals, volatilities, correlation, and horizon; equal foreign-currency amounts do not imply equal risk.

← Back to Main Slide

Appendix: Forward Contract Settlement & Credit Risk

Forward Contract Mechanics:

- Over-the-counter (OTC) customized contract negotiated directly with a bank dealer.
- Standard settlement: Deliver base currency $\mathcal{F}$ and receive quote currency H at agreed rate F_0 .
- Non-Deliverable Forwards (NDF): Common for currencies with onshore convertibility or deliverability restrictions (e.g., INR and KRW). They are cash-settled in an agreed settlement currency against a specified fixing; exact settlement depends on the quotation convention.

Credit Risk & Counterparty Exposure:

- OTC forwards carry counterparty exposure; collateral, netting, clearing, and margin reduce but do not necessarily eliminate it. Exchange-traded futures also have default risk, mitigated through a clearinghouse and daily margining.
- Dealers manage exposure through credit limits, ISDA documentation, netting, and collateral terms.

← Back to Main Slide

Appendix: Algebraic Proof of MMH & CIP Equivalence

Receivable MMH Derivation: Let $R_{\mathcal{F}}$ be the foreign receivable due at date T .

- ① Borrow foreign currency today: $B_{\mathcal{F}} = \frac{R_{\mathcal{F}}}{1+i_{\mathcal{F}}t}$.
- ② Convert to home currency at spot $S_{\mathcal{F}/\text{H},0}$: Amount in H = $\frac{R_{\mathcal{F}}}{1+i_{\mathcal{F}}t} \cdot S_{\mathcal{F}/\text{H},0}$.
- ③ Invest in home currency at rate i_{H} until T :

$$\text{Payoff}_T = R_{\mathcal{F}} \cdot \left[S_{\mathcal{F}/\text{H},0} \frac{1+i_{\text{H}}t}{1+i_{\mathcal{F}}t} \right]$$

By Covered Interest Parity (CIP), $F_{\mathcal{F}/\text{H},0,T} = S_{\mathcal{F}/\text{H},0} \frac{1+i_{\text{H}}t}{1+i_{\mathcal{F}}t}$.

Substituting CIP into the MMH payoff yields:

$$\text{Payoff}_T^{\text{MMH}} = R_{\mathcal{F}} \cdot F_{\mathcal{F}/\text{H},0,T} = \text{Payoff}_T^{\text{Forward}} \quad \blacksquare$$

Appendix: Currency Option Pricing & Greeks

Garman-Kohlhagen Model (FX Extension of Black-Scholes): For a foreign currency option with spot S , strike K , continuously compounded domestic rate i_H , foreign rate i_F , volatility σ , and time T :

$$C = Se^{-i_F T} N(d_1) - Ke^{-i_H T} N(d_2), \quad P = Ke^{-i_H T} N(-d_2) - Se^{-i_F T} N(-d_1)$$

where $N(\cdot)$ is the standard normal CDF, $d_1 = \frac{\ln(S/K) + (i_H - i_F + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, and $d_2 = d_1 - \sigma\sqrt{T}$.

Key Option Drivers (Greeks):

- **Delta (Δ):** Sensitivity of option price to spot rate S .
- **Vega:** Holding other inputs fixed, option value rises with implied FX volatility. Premium also depends on strike, tenor, rates, liquidity, and supply/demand.

[← Back to Main Slide](#)

Appendix: Comprehensive Receivable Payoff Matrix

[Background Context]

Future Spot Rate S_T	Unhedged Revenue	Forward Revenue	MMH Revenue	Put Gross Revenue	Put Net Revenue (less Prem FV)
7.50	7,500,000	8,600,000	8,600,000	8,500,000	8,290,000
7.80	7,800,000	8,600,000	8,600,000	8,500,000	8,290,000
8.00	8,000,000	8,600,000	8,600,000	8,500,000	8,290,000
8.40	8,400,000	8,600,000	8,600,000	8,500,000	8,290,000
8.50	8,500,000	8,600,000	8,600,000	8,500,000	8,290,000
8.60	8,600,000	8,600,000	8,600,000	8,600,000	8,390,000
8.81	8,810,000	8,600,000	8,600,000	8,810,000	8,600,000 (Break-Even)
9.00	9,000,000	8,600,000	8,600,000	9,000,000	8,790,000
9.50	9,500,000	8,600,000	8,600,000	9,500,000	9,290,000

← Back to Main Slide

Appendix: Comprehensive Payable Payoff Matrix

[Background Context]

Assume a US firm owes **€1,000,000** in 1 year. Rates and premiums are USD per EUR. $S_0 = 1.10$, $F_0 = 1.12$, Call Option Strike $K = 1.10$, Premium $p = \$0.03$ (FV = \$0.0315).

Future Spot Rate S_T	Unhedged Total Cost	Forward Total Cost	Call Gross Cost	Call Net Cost (incl. Prem FV)
\$1.00	\$1,000,000	\$1,120,000	\$1,000,000	\$1,031,500
\$1.05	\$1,050,000	\$1,120,000	\$1,050,000	\$1,081,500
\$1.10	\$1,100,000	\$1,120,000	\$1,100,000	\$1,131,500
\$1.15	\$1,150,000	\$1,120,000	\$1,100,000	\$1,131,500 (Capped)
\$1.25	\$1,250,000	\$1,120,000	\$1,100,000	\$1,131,500 (Capped)

← Back to Main Slide

[Background Context]

Illustrative Corporate Governance Choices:

- **Committed exposures:** A firm may use a high hedge ratio for contracted receivables/payables when budget certainty is the priority.
- **Forecast exposures:** A layered or lower hedge ratio can reduce overhedging when timing and quantity are uncertain.

Hedge Accounting Rules (FASB ASC 815 / IFRS 9):

- **Fair Value Hedge:** Offsets changes in fair value of recognized assets/liabilities. Gains/losses reported in current earnings.
- **Cash Flow Hedge:** Offsets variability in expected future cash flows. Effective portion of derivative gain/loss is deferred in Other Comprehensive Income (OCI) until the underlying transaction impacts earnings.

← Back to Main Slide

[Background Context]

Financial hedges (forwards/options) are short-term fixes. Long-term risk management requires **operational restructuring**:

- ① **Currency Matching**: Structure supply contracts so that revenues and expenses are denominated in the same currency. (e.g. Japanese automaker buys steel in USD for US assembly plant).
- ② **Plant Relocation & Shifting Production**: Move manufacturing facilities to key export market countries to create a natural expense hedge.
- ③ **Cross-Currency Debt Financing**: Issue corporate bonds in foreign host currency where future operational cash flows are generated.

← Back to Main Slide

Appendix: Economic Exposure Case (Madison Co.)

[Background Context]

Madison Co. Setup: US manufacturer with \$320M US sales, C\$4M Canadian sales, \$50M US material costs, C\$200M Canadian material costs, \$3M USD interest expense, and C\$10M interest expense.

Item	C\$ = \$0.75	C\$ = \$0.80	C\$ = \$0.85
Total Revenue (USD)	\$323.0M	\$323.2M	\$323.4M
Total Material Costs (USD)	\$200.0M	\$210.0M	\$220.0M
Operating Exp.	\$60.0M	\$60.0M	\$60.0M
Interest Exp. (USD)	\$10.5M	\$11.0M	\$11.5M
Net Operating Cash Flow	\$52.5M	\$42.2M	\$31.9M

- **Result:** As Canadian Dollar appreciates ($0.75 \rightarrow 0.85$), Madison's USD cash flow drops by 39% because Canadian cost outflows far exceed Canadian revenue inflows.

← Back to Main Slide

Functional Currency & Translation Methods:

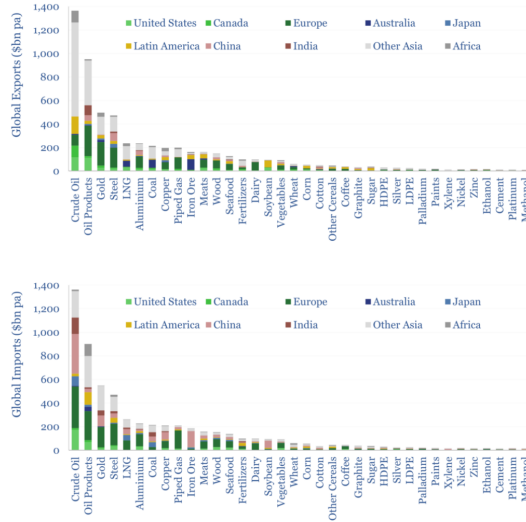
- **Translation / Current-Rate Method** (when the foreign entity's functional currency differs from the parent's reporting currency):
 - Assets and Liabilities translated at current spot rate on balance sheet date.
 - Income Statement items translated at weighted average exchange rate for the period.
 - Translation gains/losses bypass income statement and go directly to balance sheet equity (CTA - Cumulative Translation Adjustment).
- **Remeasurement / Temporal Method** (when the entity's books are not kept in its functional currency; under US GAAP a highly inflationary foreign economy generally uses the reporting currency as functional currency):
 - Monetary assets/liabilities translated at current rate; non-monetary items (inventory, land) translated at historical rate.
 - Remeasurement gains/losses generally enter current net income. IFRS treatment of hyperinflationary operations differs and uses IAS 29 together with IAS 21.

Appendix: Commodities, forwards, and futures

- Commodity contracts standardize grade, quantity, delivery location, and date. Products may still differ in quality, so a benchmark hedge leaves basis risk.
- A forward is a bilateral future purchase or sale. Exchange-traded futures standardize terms and usually require initial margin and daily variation margin.
- A farmer shorts futures against expected harvest revenue. A processor may buy futures against future input costs. Crop failure can leave the farmer overhedged.
- With storage cost rate u and convenience yield y , a stylized carry relation is $F_0 = S_0 e^{(r+u-y)T}$. Storage and short-sale constraints can limit arbitrage.

← Back to Main Slide

Commodities Trade



Appendix: A farmer's forward and put hedge

Assume a certain harvest of 10 units, debt repayment USD 70, forward price USD 9, and a put with strike USD 8.50 and premium USD 1.50 per unit. Ignore premium financing.

Harvest price	Forward profit	Put-hedged profit
USD 6	$90 - 70 = 20$	$85 - 70 - 15 = 0$
USD 9	$90 - 70 = 20$	$90 - 70 - 15 = 5$
USD 12	$90 - 70 = 20$	$120 - 70 - 15 = 35$

- The forward locks profit at USD 20. The put gives a zero-profit floor under the stated quantity and financing assumptions.
- An option premium purchases flexibility. It must be included even when the option expires unused.

Incentives for Options Contracts

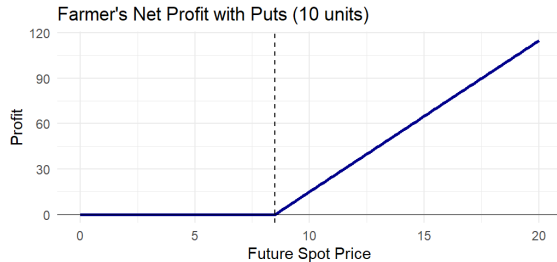
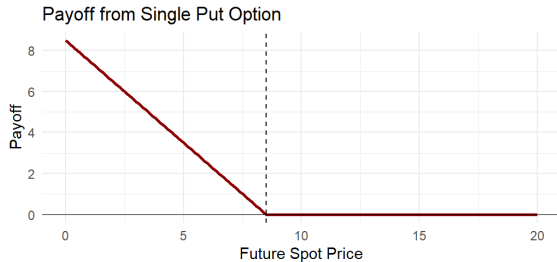


Illustration: strike 8.50, premium 1.50 per unit, harvest 10, and debt 70. Include the premium in net profit.

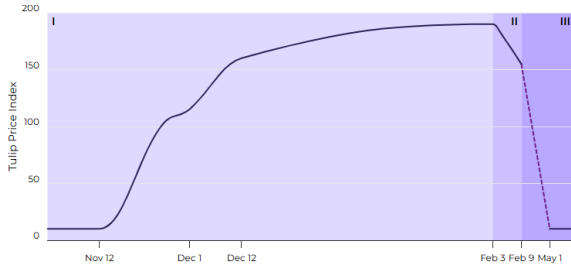
[← Back to Main Slide](#)

Appendix: Early commodity and securities markets

- Tulip contracts in the 1630s illustrate speculative contracting and counterparty enforcement risk. Claims about a purely irrational bubble remain historically contested.
- The Dojima rice market in Osaka developed standardized rice claims and futures trading. Warehouse receipts tied financial claims to specified physical goods.
- Antwerp and Amsterdam connected merchant finance and early derivatives. Chicago later institutionalized standardized agricultural futures, and CBOE launched listed equity options in 1973.
- The Dutch East India Company (VOC), chartered in 1602, combined shareholder capital, borrowing, supplier advances, and credit to auction buyers within a colonial trading enterprise.

← Back to Main Slide

Example: Tulip Mania in the Netherlands



← Back to Main Slide



The wonderful depiction of a Dutch tulip cultivation: Johannes Boschaert's 'Still Life with Tulips' from 1638 hangs at the National Museum in Stockholm. Photo: National Museum

Example: Tulip Mania in the Netherlands



← Back to Main Slide